Hong Kyu Lee

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LinkedIn | GitHub | Website

Ph.D. Candidate in Trustworthy AI, Machine Unlearning, and Privacy

Research Profile Ph.D. candidate at Emory University researching machine unlearning, LLM/LVLM optimization, AI security, and privacy-preserving learning. Published in Findings of the Association for Computational Linguistics (ACL) 2026, IJCAI 2025, ACM WWW 2024, IEEE ICC 2021, and Computers & Security 2021.

Machine Unlearning **LLM Agents** **Privacy Attacks** **Differential Privacy** **Graph ML**

Selected Publications

Direct Token Optimization: A Self-Contained Approach to Large Language Model Unlearning
Hong Kyu Lee, Ruixuan Liu, Li Xiong
Findings of the Association for Computational Linguistics (ACL), 2026, paper link

Contrastive Unlearning: A Contrastive Approach to Machine Unlearning
Hong Kyu Lee, Qiuchen Zhang, Jian Lou, Carl Yang, Li Xiong
International Joint Conference on Artificial Intelligence (IJCAI 2025), paper link

DPAR: Decoupled Graph Neural Networks with Node-Level Differential Privacy
Qiuchen Zhang, **Hong Kyu Lee**, Jian Lou, Carl Yang, Li Xiong
ACM Web Conference (WWW 2024), paper link

Research Experience

Research Intern **Samsung Research America** **Mountain View, CA**
May 2026 – Aug 2026

Memory Systems for Implicit Reasoning Agents | Python, OpenAI API, Docker, vLLM

- Developing an episodic-memory retrieval model for long-horizon implicit reasoning agents, with model-serving and evaluation workflows built in Python, Docker, and vLLM.

Doctoral Student **Emory University** **Atlanta, GA**
Aug 2022 – Present

Advised by Prof. Li Xiong and Prof. Carl Yang

LVLM Unlearning | PyTorch, Hugging Face, Transformers

- Developed a vision-token-guided unlearning method for LLaVA and Qwen2-VL that removes cross-modal and text-only knowledge while preserving multimodal utility.
- Achieved up to 7× higher forget quality than DPO, NPO, and MANU baselines on CLEAR and MLLMU.

Membership Inference Attacks on Vision-Language Models | PyTorch, Hugging Face

- Designed a black-box membership inference attack for LLaVA and MiniGPT-4 by perturbing input images and using response changes as membership signals.
- Improved AUC by up to 18% over the VL-MIA baseline on Flickr and COCO-2017.

Resource-Free LLM Unlearning | PyTorch, Hugging Face

- Developed a self-contained LLM unlearning framework requiring only the target model and forget set, without retain data, reference models, external classifiers, or API models.
- Increased TOFU forget quality from 0.05 to 0.91 compared with TPO and FLAT baselines, with evaluation on TOFU and MUSE.

Feature Unlearning via Sharpness-Aware Selection | PyTorch

- Developed a saliency-based parameter selection method for feature-level unlearning, targeting tabular-style data features in GPT-2 and ResNet models.

Graph Neural Network Unlearning | PyTorch, PyTorch Geometric, DGL

- Designed Node-CUL, a node-level GNN unlearning method validated through representation alignment on Cora, PubMed, and CS.
- Outperformed GNNDelete and GIF in runtime while preserving node-classification accuracy.

Contrastive Classifier Unlearning | PyTorch, CLIP

- Proposed a contrastive unlearning framework for both classification models and embedding models, including ResNet, ViT, and CLIP.
- Improved test accuracy by 5% and reduced runtime by 20% over Gradient Ascent and SCRUB on Mini-ImageNet, CIFAR-10, and SVHN.

Differentially Private Graph Neural Networks | PyTorch, TensorFlow

- Helped design and evaluate a PageRank-based node-level differentially private GNN, improving utility from 0.705 to 0.934 over GAP at the same privacy level on CoraML, PubMed, and CS.

Earlier Publications

On Defensive Neural Network Against Inference Attack in Federated Learning
Hong Kyu Lee, Jeehyeong Kim, Rasheed Hussain, Sunghyun Cho, Junggab Son
IEEE International Conference on Communications (ICC 2021), paper link

Digestive Neural Networks: A Novel Defense Strategy Against Inference Attacks in Federated Learning
Hong Kyu Lee, Jeehyeong Kim, Seyong Ahn, Rasheed Hussain, Sunghyun Cho, Junggab Son
Computers & Security 2021, paper link

Efficient yet Robust Privacy Preservation for MPEG-DASH-Based Video Streaming
Luke Cranfill, Jeehyeong Kim, **Hong Kyu Lee**, Victor Youdom Kemmoe, Sunghyun Cho, Junggab Son
Security and Communication Networks 2021, paper link

Preprints

Sharpness-Aware Parameter Selection for Machine Unlearning
Saber Malekmohammadi, **Hong Kyu Lee**, Li Xiong
arXiv:2504.06398, 2025, preprint

Node-Level Contrastive Unlearning on Graph Neural Networks
Hong Kyu Lee, Qiuchen Zhang, Carl Yang, Li Xiong
arXiv:2503.02959, 2025, preprint

Additional Experience

Software Engineer **Telchemy Inc.** **Alpharetta, GA**
Sep 2021 – Jun 2022

- Implemented a low-latency C clustering module for real-time network packet analysis on Linux systems.
- Met latency and performance constraints for production network monitoring workflows.

Graduate Research Assistant **Kennesaw State University** **Kennesaw, GA**
Aug 2019 – May 2021

- Built privacy-preserving federated learning defenses for secure IoT platforms and sensitive prediction tasks.
- Evaluated privacy-preserving algorithms for MPEG-DASH video streaming under multiple threat models.

Education

Emory University **Ph.D. in Computer Science, GPA: 3.96** Atlanta, GA; Aug 2022 – Present

Anticipated graduation: May 2027
Kennesaw State University **M.S. in Computer Science, GPA: 4.0** Kennesaw, GA; Aug 2019 – May 2021
Hanyang University ERICA **B.S. in Computer Science, GPA: 3.81** Ansan, South Korea; Mar 2015 – Aug 2019

Technical Skills

Languages:	Python, C, C++
ML:	PyTorch, Hugging Face, Transformers, TensorFlow, scikit-learn, LLaVA, LLaMA, ViT, CLIP
Graph ML:	PyTorch Geometric, DGL, NetworkX
Systems:	Linux, Slurm, Docker, Git, GitHub